

CHEMISTRY | UNIT 02 | LESSON 04

Electron Arrangement and Periodic Position

Notebook Evidence - Independent Application

Student name:		Date:	
MLA standard:	MLA.CHEM.ATO.03	Submission:	Notebook Evidence - Independent Application

Directions

Use the PhET Build an Atom simulation linked in Moodle. Complete each trial, then use the electron arrangement to identify the period and valence-electron pattern. Submit this PDF in U2 L4 Notebook Evidence Submission.

Part A - Simulation evidence

Trial	p+	n0	e-	Element	Charge	Electron arrangement
1						
2						
3						

Required atoms: Trial 1 = 6 p+, 6 n0, 6 e-; Trial 2 = 11 p+, 12 n0, 11 e-; Trial 3 = 12 p+, 12 n0, 10 e-.

Part B - Connect arrangement to periodic position

Trial	Occupied levels / period	Outer electrons	Periodic-position conclusion
1			
2			
3			

Explain why Trial 2 and Trial 3 have different chemical behavior even though their nuclei differ by only one proton.
